

Predictive Maintenance Dashboard

Client Requirements for Full Production Deployment

Prepared: June 30, 2026 · OpsFlo Engineering

The dashboard is fully built and running on a 10,000-asset synthetic dataset. The items below are the only outstanding requirements needed from the client to connect it to real fleet data, secure it, and deploy it to a production environment. Items marked **[NEW]** were identified during the final technical audit of the codebase.

A Data & Telemetry Requirements 1 – 5

#1 — Live Telematics API Access

API credentials or a scheduled data export connection from the client's telematics platform.

The following sensor fields must be present in the data feed:

- Engine: ECT (coolant temp), OILP_press (oil pressure), RPM, eng_load
- Generator output: ekWattsAvg (kW), eGenVoltage A/B/C, eGenCurrent A/B/C, eGenFrequency, ePFAvg
- Intake / manifold: AAT, IAT, TIP, MAP, NG Fuel Press, EPR
- Spark / throttle: spk_adv, KNK_retard, TPS pct, FuelPercents
- System: FaultSysFlag, dwell1, dwell2, HM_hours (runtime), Latitude, Longitude, timestamp

#2 — Column / Field Name Mapping **[NEW]**

The client's telematics system will use different internal field names to ours.

Required: a mapping document from their field names to the expected column names above.

Example: their system may export coolant_temp_c where we expect ECT.

Without this, no sensor data will flow into the ML engine correctly.

#3 — Data Refresh Cadence **[NEW]**

The dashboard currently loads data once at startup and never refreshes.

Client must specify the desired live update frequency: every 5 min / 15 min / 1 hour.

This determines the polling or streaming architecture we build for production.

#4 — Historical Telemetry Data

Minimum 30 days of history per asset.

Required to compute the 1hr / 6hr / 24hr rolling window features that power all ML models.

Without history, ML predictions fall back to rule-based logic only.

#5 — Historical Maintenance & Failure Records

Past failure events per asset: asset ID, timestamp, failure type.

Work order history: scheduled vs unscheduled maintenance events.

These records are used to generate training labels (failure_label, failure_scenario) for the ML models.

Without labeled history the models cannot be retrained on real failure patterns.

B

Fleet & Asset Information

Requirements 6 – 9

#6 — Asset Registry & Naming Convention

Full list of real asset IDs with human-readable display names.

Currently the dashboard auto-generates placeholder names: NGG-1001, NGG-1002, ...

Required: the client's actual naming/numbering format and asset list.

#7 — Rated (Nameplate) Capacity per Asset [\[NEW\]](#)

Each asset card displays a kW capacity figure.

Currently this incorrectly uses the live kW output reading rather than rated capacity.

Required: nameplate/rated kW per asset ID from the asset registry.

#8 — Fuel Type per Asset [\[NEW\]](#)

The telemetry panel shows fuel type for each asset.

Currently defaults to 'Natural Gas' for all assets when the field is missing.

Required: fuel type per asset (Natural Gas / Diesel / Propane / Bi-Fuel / Other).

#9 — Fleet Size & Site Structure [\[NEW\]](#)

Total number of assets in the live fleet (for server sizing — 10k assets takes ~40s startup).

Is this a single site or multiple sites / regions?

Currently there is no multi-fleet or multi-site concept in the dashboard.

If multi-site is needed, scope must be agreed before deployment.

C

Business Rules & Configuration

Requirements 10 – 12

#10 — Risk Threshold Approval [\[NEW\]](#)

Current thresholds (health score → risk level):

• 0–24: Critical • 25–49: High • 50–74: Medium • 75–100: Low

Client must review and approve these thresholds or provide revised values.

#11 — Failure Scenario Sign-off **[NEW]**

The dashboard has 11 pre-defined failure scenarios with penalty weights:

Overheating, Lubrication Failure, Complete Shutdown, Electrical Failure, Sensor/Comm Fault, Partial Load Failure, Cascading Failure, Fuel Starvation, Cooling Partial, Voltage Fluctuation, Normal.

Client must confirm these match their real failure modes, flag any that don't apply, and provide any additional scenarios not listed.

#12 — PM Schedule Baseline

The PM Job Optimization panel currently uses a hardcoded job name ('500 HR Service') and a fixed baseline date.

Required from client:

- Their actual PM job names and service intervals per asset type
- Current scheduled due dates per asset so the AI adjustment works on real jobs

D

Infrastructure & Deployment

Requirements 13 – 14

#13 — Hosting Environment

A server or cloud VM to deploy to: AWS / Azure / GCP / on-prem.

Minimum spec: 2 vCPU, 4 GB RAM (all 7 ML models load into memory at startup).

A domain or subdomain for the dashboard URL.

An SSL certificate (HTTPS is required for any production deployment).

#14 — Authentication & Access Control

The dashboard currently has zero authentication — anyone with the URL can see all fleet data.

Required: list of users who need access, with their roles:

- Admin / Fleet Manager / Technician / Read-only

Preferred auth method: SSO/SAML (company directory), Microsoft/Google OAuth, or username + password.

E

Integrations & Notifications

Requirements 15 – 16

#15 — CMMS / Work Order System Integration

The 'Create Work Order' button currently returns a mock success message — no real WO is created.

Required: API access to the client's CMMS (SAP PM / IBM Maximo / Fleetio / UpKeep / other) so accepted recommendations generate real work orders.

Provide: CMMS name, API endpoint, and credentials.

#16 — Alert Channels & On-call Contacts

Specify where critical alerts should be sent: Email / SMS / Slack / Microsoft Teams / PagerDuty.

Define the risk score threshold that triggers an alert.

Provide on-call contacts per asset group or site.

F

Compliance & Data Governance

Requirement 17

#17 — Data Retention, Residency & PII Policy

How long should telemetry data be retained on the server?

Any regional data storage requirements (data must stay in a specific country/region)?

The dataset includes a Driver name field — a PII handling policy is required before go-live.



Priority Summary

All 17 requirements at a glance

#	Requirement	Section	Priority	Blocks If Missing
1	Live Telematics API Access	A	Must have	Everything
2	Column / Field Name Mapping	A	Must have	Data ingestion
3	Data Refresh Cadence	A	Must have	Live updates
4	Historical Telemetry Data	A	Must have	ML rolling features
5	Historical Maintenance & Failure Records	A	Must have	ML retraining
6	Asset Registry & Naming Convention	B	Must have	Asset display
7	Rated Capacity per Asset	B	Must have	Correct asset cards
13	Hosting Environment	D	Must have	Deployment
14	Authentication & Access Control	D	Must have	Security
8	Fuel Type per Asset	B	Medium	Telemetry display
9	Fleet Size & Site Structure	B	Medium	Server sizing
10	Risk Threshold Approval	C	High	Prediction accuracy
11	Failure Scenario Sign-off	C	High	Prediction accuracy
12	PM Schedule Baseline	C	High	PM optimization
15	CMMS / Work Order Integration	E	High	Work order flow
16	Alert Channels & On-call Contacts	E	High	Operational alerts
17	Data Retention, Residency & PII	F	Medium	Compliance

Items marked [NEW] were identified during the final technical audit of the codebase and were not in the initial requirements pass.